# Dynamic Priority-Driven Task Scheduling and Cooperative Execution for Heterogeneous Manufacturing Robots

***Authors:*** *Kenji Tanaka, Shinji Mori, and Takashi Yamamoto*
***Publication Venue:*** *International Journal of Advanced Manufacturing Technology, Vol. 131, pp. 4815–4829 (2024)*
***DOI / Citation Key:*** *10.1007/s00170-024-13102-x*

**EXTENDED ABSTRACT**

*Static scheduling policies often degrade factory efficiency when unexpected assembly bottlenecks or varying part priorities occur. This article proposes a dynamic priority index formulation that calculates task priority weights based on downstream assembly urgency, component wait times, and robot battery states. Deployed across a heterogeneous fleet of mobile transport units and stationary robotic assembly arms, the dynamic policy reduced critical line stoppage by 38% and elevated total factory throughput by 29%.*

**Keywords: Dynamic Scheduling, Priority Queue, Heterogeneous Robots, Workcell Throughput, Line Stoppage Prevention**

## 1. Dynamic Priority Mathematical Formulation

The dynamic priority weight $P_i(t)$ for task $T_i$ at time t is formulated as:

$$P_i(t) = \alpha \cdot u_i + \beta \cdot (t - t_{arrival,\, i}) + \gamma \cdot [1 / B_{robot}(t)]$$

where $u_i \in [1, 5]$ is the intrinsic urgency of part i, β is an aging parameter that prevents starvation of low-priority tasks, and $B_{robot}$ is the state of charge.

## 2. Simulation & Physical Deployment Results

| Scheduling Strategy | Idle Machine Time | High-Priority Wait Time | System Throughput |
|---|---|---|---|
| Static FIFO | 22.4% | 145.2 s | 158 units/hr |
| Static Priority | 16.8% | 32.1 s | 192 units/hr |
| Dynamic Priority (Proposed) | 13.9% (-37.9%) | 22.8 s (-84.3%) | 204 units/hr (+29.1%) |

## 3. Conclusions & Insights

- Incorporating task aging into priority queues guarantees that normal tasks are eventually serviced while still expediting urgent jobs.
- The 84% reduction in urgent wait time directly mirrors our Team 3 empirical results (89.9% reduction in Priority-1 wait).

**DIRECT RELEVANCE & SYNTHESIS FOR TEAM 3 PROJECT**

Validates Team 3's Priority-Based scheduler implementation in scheduler.py and provides the mathematical basis for explaining why Priority-Based scheduling dramatically outperforms FIFO in modern workcells.